

Mathematics A

Unit 2 • Chapter OL-T47 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

5 classified items • 22 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T47 • Expertise Q16+ • 5 items • 22 marks

Chapter	Items	Marks	Starts
01. OL-T47 Area & Volume of Similar Shapes	5	22	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Area & Volume of Similar Shapes

Unit 2 • Expertise • 22 marks • 5 item(s)

June 2025 | 4MA1-2H Q26 | 4 marks

Solve a volume difference or sum for similar solids

November 2023 | 4MA1-1H Q23 | 5 marks

Convert between area and volume ratios

November 2023 | 4MA1-2H Q23 | 5 marks

Use similar solids to solve a frustum problem

January 2022 | 4MA1-1H Q17 | 4 marks

Solve a volume difference or sum for similar solids

January 2023 | 2H Q19 | 4 marks

Solve an area difference or sum for similar figures

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

The height of solid **A** is 31 cm
The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm³

(Total for Question 26 is 4 marks)

WORKING SPACE

4MA1-1H Q23 demand

5 marks

23 Solid A is similar to solid B

Here is some information about solid A and solid B

	solid A	solid B
Height (cm)	3^x	
Area (cm ²)	7776	486
Volume (cm ³)	8^x	2^{x+4}

Work out the height of solid B
Give your answer as a decimal.

..... cm

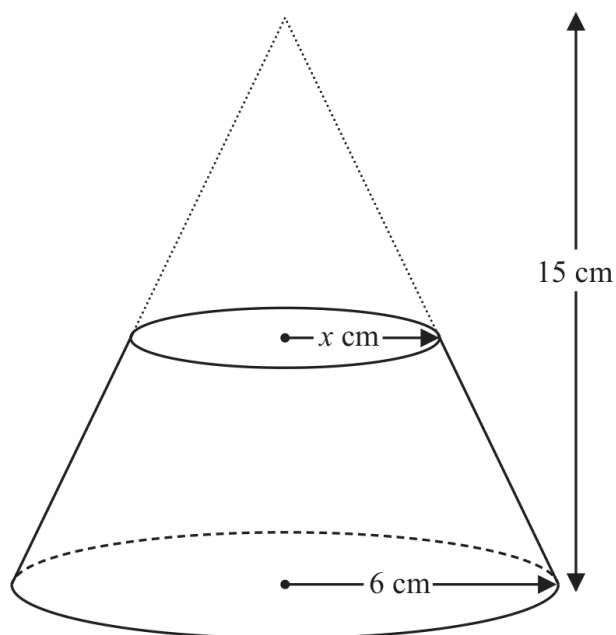
WORKING SPACE

4MA1-2H Q23 whole

5 marks

23 Here is a frustum of a cone.

The frustum is made by removing a small cone from a similar large cone.



The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.

The radius of the base of the small cone is x cm.

Given that the volume of the frustum is $\frac{4212}{25}\pi \text{ cm}^3$

work out the value of x

Show clear algebraic working.

$x = \dots\dots\dots$

WORKING SPACE

Continue your method

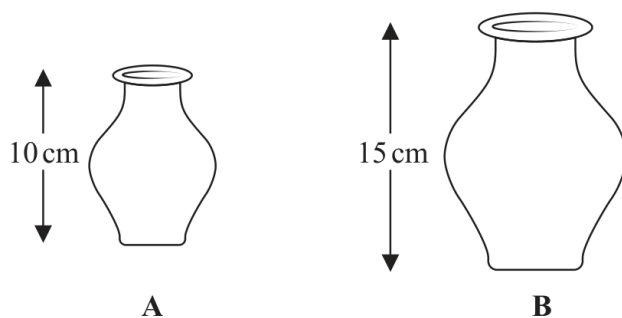
5 marks

WORKING SPACE

4MA1-1H Q17

4 marks

17 **A** and **B** are two similar vases.



Vase **A** has height 10 cm.

Vase **B** has height 15 cm.

The difference between the volume of vase **A** and the volume of vase **B** is 1197 cm^3

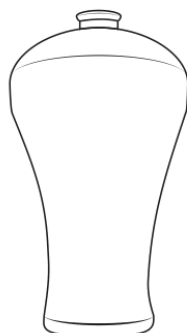
Calculate the volume of vase **A**

..... cm^3

Continue your method**4 marks**

WORKING SPACE

19 **A** and **B** are two similar vases.



A



B

Diagram **NOT**
accurately drawn

The vases are such that

$$\text{surface area of vase B} = \frac{25}{64} \times \text{surface area of vase A}$$

and that

$$\text{volume of vase A} - \text{volume of vase B} = 541.8 \text{ cm}^3$$

Calculate the volume of vase **B**

..... cm³

(Total for Question 19 is 4 marks)

YOUR WORKING
